Program	B. Tech CSE: All Batches	Semester: I
Course	Physics for Computer Engineers	Course Code : PHYS 1036
Session	July-December, 2025	Unit I: Laser & Fibre Optics

Q. No.	Unit I @ Laser and Fibre Optics	CO
1	Distinguish between ordinary light and laser light.	1
	Distinguish between spontaneous and stimulated emissions.	1
2	Write Einstein's coefficients for absorption, spontaneous emission and stimulated emission. Derive a relation between them.	1
3	Show that the ratio of spontaneous to stimulated emissions is propotional to the cube of the incident frequency.	1
4	Explain the term population inversion in connection with the laser. Name different types of pumping mechanism?	1
5	What are the basic requirements for producing laser? Explain	1
6	Explain the construction & working of Ruby laser/pulsed laser/three level laser with suitable energy level diagram.	1
	With the help of suitable energy level diagram explain the working of a three-level laser?	
7	Explain the construction & working of He-Ne/gas laser/continuous laser/four level laser with suitable energy level diagram.	1
	With the help of suitable energy level diagram, explain the working of a four-level Laser?	
8	Write various applications of laser in different fields.	1
9	Distinguish between photography and holography.	1
	Compare the information stored in a hologram with that in a photograph.	
10	What is holography? What are the fundamental requirements to construct a hologram?	1
11	Discuss the construction & re-construction of a hologram with suitable diagram. Write the applications of holography.	1
	What is the principle used in holography? Explain the method by which image is obtained from a hologram?	
12	Mention the advantages of optical fibres in communication system.	1
13	What is an optical fiber? Draw the structure of an optical fiber and explain.	1
14	What is the basic principle in transmission of a signal through an optical fibre.	1
15	Derive the expression for Numerical Aperture (NA) of an optical fibre. Also state the physical significance of the NA.	1
16	What are the types of optical fibers? Compare single mode, multimode step index and graded index fiber with the schematic diagrams.	1
17	Explain the attenuation in optical fibre. What are the important factors responsible for the loss in optical fiber? Discuss.	1
18	Drive a relation for the attenuation coefficient in an optical fiber.	1
19	Explain optical fibre communication system with the block diagrams.	1
	An optic fiber of refractive index 1.50 is to be clad to ensure total internal	

	reflection that will contain light travelling with in 5^0 of the fiber axis. What minimum refractive index is allowed for the cladding? NA = Sin(5 degree) = $((1.5)^2 - (n_2)^2)^{1/2}$ $n_2 = 1.497$	

NOTE: For numericals, refer the tutorial sheet and any of the text/reference books.